

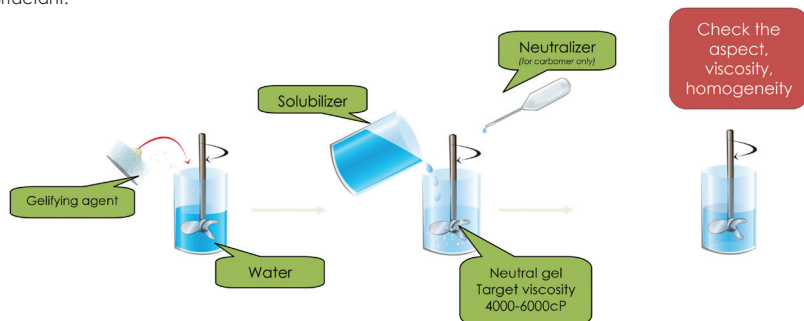
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INTRODUCTION

The development of efficient dosage form for skin delivery requires the use of solubilizers that improve drug solubility and penetration through the skin. These solubilizers are generally lipophilic by nature. Only few of them are miscible in aqueous phase and therefore are generally added in oily or surfactant phase. The aim of the study was to formulate some of these lipophilic solubilizers in aqueous gels.

EXPERIMENTAL METHODS

Different solubilizers from Gattefossé were used in the study: Transcutol® P (co-solvent), Labrasol® and Labrafil® M1944CS (water-dispersible surfactants), Plurao® oleique CC497 and Capryol™ 90 (water-insoluble surfactants). Three gelling agents were compared: hydroxyethylcellulose (HEC), hydroxypropylcellulose (HPC) and carbomer 980. The initial target viscosity of the gel is 4,000-6,000 cP. The corresponding amounts of gelling agent are: 2% HEC, 2.5% HPC and 0.5% carbomer 980. Binary diagrams were designed to establish the maximum amount of solubilizers that can be added into aqueous gels. Ternary diagrams were then designed with the addition of a surfactant.



RESULTS AND DISCUSSION

Transcutol® P is a co-solvent therefore can be added in aqueous gels up to 50% (Figure 1a). Viscosity and clearness is however reduced above 30% of Transcutol® P, except for HPC gel. Labrasol® behaves in gels like in water phase: below 2% it makes white gels but above 2%, it makes microemulsion and gels remain clear (Figure 1b). The maximum amount of Labrasol® in gels depends on the gelling agent: 40% in carbomer gels but only 5% in HEC or HPC gels. Labrafil® M1944CS (Figure 1c) and Plurao® oleique CC497 (Figure 1d) generate white gels with a maximum of 10% solubilizer in carbomer 980 gel, 6% in HEC gel or 5% in HPC gel.

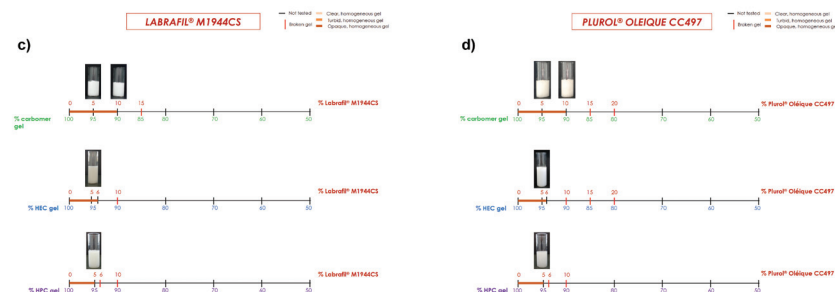
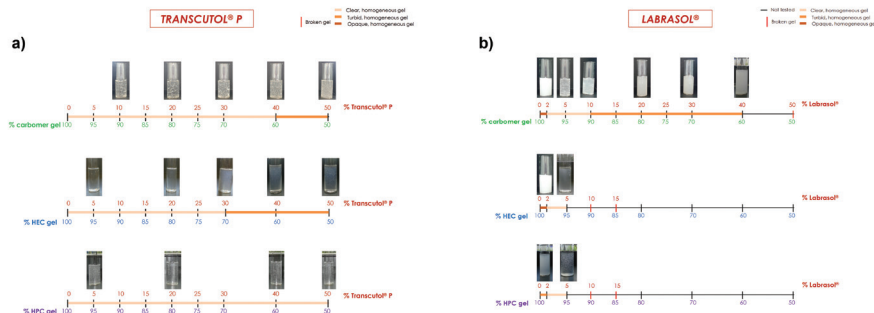


Figure 1: Binary diagrams of: a) Transcutol®, b) Labrasol®, c) Labrafil M1944CS® and d) Plurao® oleique® CC497, with carbomer 980, HEC and HPC gels.

Capryol™ 90 can only be dispersed in HPC gel and makes white gels with 6% maximum of solubilizer (Figure 2a). For HEC or carbomer 980 gels, Capryol™ 90 cannot be dispersed and a surfactant is required. White gels are homogeneous up to 20% of Capryol™ 90 / Tween® 80 (ratio 3:1) (Figure 2b).

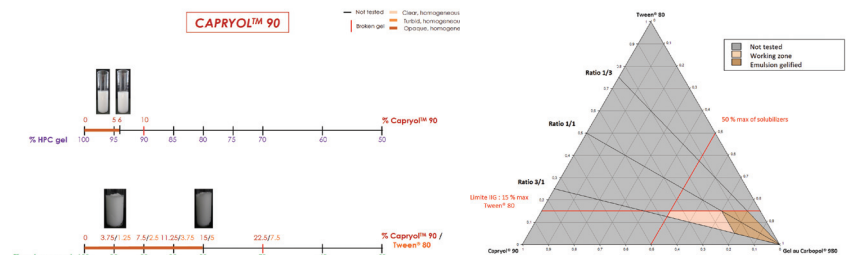


Figure 2: Capryol™ 90: a) binary diagrams of Capryol™ 90 with HPC and carbomer 980 gels and b) ternary diagram of Capryol™ 90 / Tween® 80 with carbomer 980 gel.

CONCLUSION

Depending on their affinity for water phase, dispersibility of solubilizers in aqueous gels can be limited (Figure 3). The addition of surfactant can increase their level of use. Gels are generally white in appearance with a creamy texture.

	HEC (2% Natrosol® HHX Pharm)		HPC (2.5% Klucel® HF Pharm)		Carbomer (0.5% Carbopol® 980 NF)	
	clear gel	turbid or opaque gel	clear gel	turbid or opaque gel	clear gel	turbid or opaque gel
Transcutol® P	< 30%	30-50%	≤ 50%	-	< 40%	40-50%
Labrasol®	2-5%	≤ 2%	2-5%	≤ 2%	2-10%	≤ 2% 10-40%
Labrafil® M1944CS	-	≤ 6%	-	≤ 5%	-	≤ 10%
Plurol® oleique CC497	-	≤ 6%	-	≤ 5%	-	≤ 10%
Capryol™ 90	-	-	-	≤ 6%	-	-
Capryol™ 90 / Tween® 80 (3:1)	not tested		not tested		-	≤ 20%

Figure 3: Ranges of solubilizers by gelling agent.